

Education

The University of Hong Kong

Sep 2023 – Jun 2027 (expected)

B.Sc. in Mathematics & Computer Science (Minor)

- CGPA: 3.83/4.3 | **Dean's Honours List** (2023-2024 & 2024-2025)
- Courses: Analysis, Algebra, Probability, Algorithms, Optimization, Discrete mathematics, Robotics, Computer Vision

National University of Singapore

Jan 2026 – May 2026

- HKU Worldwide Exchange Program | **Tang Pak Kau Scholarships**

Research Interests

- Statistical representation learning · Optimization theory for machine learning · Next-generation AI · Reliable & Interpretable AI · AI for social impact

Publications and Awards

- Liu, W.*, **Liu, H.***, et al. (2026) . *From Backward Spreading to Forward Replay: Revisiting Target Construction in LLM Parameter Editing*. **Accepted by International Conference on Machine Learning 2026**.
- Liu, W., Xu, H., **Liu, H.**, et al. (2026). *Are We Evaluating the Specificity of LLM Model Editing Properly?* **Under review at conference**.
- Sungarda, J.*, **Liu, H.***, et al. (2025). *GenAI for Social Work Field Education: Client Simulation with Real-Time Feedback*. **Publication on IEEE International Conference on Big Data 2025**.
- **iCANX Innovation Award 2025** (Oct 2025)

Research Experiences

Student Research Assistant, School of Computing, NUS

Jan – May 2026

Supervisor: Prof. Lee Wee Sun

- Investigated parameter editing in LLMs; implemented and benchmarked 5+ editing baselines (ROME, MEMIT, MEND) under a unified PyTorch framework.
- Proposed a novel target construction strategy that improved editing success rate on CounterFact+ dataset while maintaining locality.
- Co-authored ICML 2026 paper; responsible for all experimental pipelines and ablation studies.

Student Research Assistant, Innovation Wing, Faculty of Engineering, HKU

Oct 2024 – present

Supervisor: Prof. Kao, Benjamin C.M. | Project: SWITCH- LLM for Social Work and Education

- Led an interdisciplinary team to design and deploy an LLM-based client simulation system for social work training; co-authored a peer-reviewed paper on the system architecture and real-time feedback mechanism (IEEE Big Data 2025).
- Fine-tuned LLM on 10k+ anonymized dialogue transcripts and systematically evaluated model via multi-dimensional metrics, conducting studies on prompt strategies to optimize trade-offs between response quality and inference latency.
- Developed a full-stack web application using Next.js with flask backend. The platform has been piloted in local NGOs and was featured in the university's annual report (2025–2026) as a signature AI-for-social-good project.
- Presented the project to university leadership, provincial officials, and visiting scholars during high-level delegation visits; served as the primary student spokesperson for system demonstrations and Q&A sessions.

Summer Research Fellowship Program, Faculty of Science, HKU

Jun – Sep 2025

Supervisor: Prof. Yunwen Lei

- Derive practical analysis on the generalization capability of current popular SGD optimization method.
- Implemented synthetic experiments in Python to validate theory; results presented in a LaTeX technical report.

Capstone Project, Faculty of Science, HKU

Aug 2026 – Jun 2027

Supervisor: Prof. Lexiao Lai

- Working on tame geometry and optimization to study non-convex optimization on specific data structures.
- Aiming to understand geometric structures of AI data representations and produce a manuscript draft for top venue.

Student Teaching Assistant, School of Computing and Data Science, HKU

Sep – Dec 2025

- Led weekly tutorials for Intro to Programming and Data Structures, serving 30+ students.

Skills

Python, C++, LaTeX, Git | Mandarin (native), English (IELTS 8.0), Cantonese (native).